Stochastic Optimization Practice
Food Bank Inventory Management

2025-2026

1 Introduction

A food bank is an emergency feeding organization that provides hunger relief to families living in poverty. We consider a food bank with 2 warehouses, denoted as $\mathcal{I} = \{1, 2\}$, that prepare food weekly for 6 districts, represented by $\mathcal{J} = \{1, 2, \ldots, 6\}$. The food demand, denoted by the random variable $\xi_{d,j}$, exhibits both spatial dependency on $j \in \mathcal{J}$ and temporal dependency on day $d \in \mathcal{D} = \{1, 2, ..., 7\}$ (Monday to Sunday) of a given week.

The decision-maker must determine the inventory amount for each warehouse, denoted by x_i, on Sunday night. The objective is to optimize the weekly ordering amount x_i and make daily planning decisions while satisfying operational constraints. To start with, there are several economic factors:

- The food bank determines the high-level decisions x_1 and x_2 every Sunday night.
- Ordering incurs an cost c_i^o per meal, which differs for each warehouse.
- Any leftover at the end of the week will spoil, and the food bank must pay a waste disposal cost c^w per meal. This quantity c^w is the same for each warehouse.
- Since the order is made on Sunday night and the leftover is disposed before the next ordering time, inventory stored at the two warehouses incurs a holding cost for the nights of Sunday (right after the ordering), Monday,. . . , and Saturday. No holding cost on next Sunday night since the leftover will be disposed. The holding cost coefficient is c^h per meal, same for each warehouse.

When determining the order quantities x_1 and x_2, the food bank also needs to consider the following operational constraints and costs:

- An organization will publish a weekly quota plan for each district after x_1 and x_2 are determined. In other words, the demand $\xi_{d,j}$ for a whole week will be revealed.
- With the published quota, along with the determined x_1 and x_2, the food bank will determine the daily supply amount $y_{d,i,j}$ at once.

- If demand $\xi_{d,j}$ is higher than $y_{d,i,j}$, according to the regulation, a penalty c^p per meal will be considered for the unmet demand $\xi_{d,j} - y_{d,i,j}$. If demand $\xi_{d,j}$ is lower than $y_{d,i,j}$, the excess amount $y_{d,i,j} - \xi_{d,j}$ will be sent back to the warehouse at night first (incurs holding cost), then be sent to the same district the next day, if d does not represent Sunday.
- To transport meal $y_{d,i,j}$ from warehouse i to j, a transportation cost $c^t_{i,j}$ per meal is considered every day. Note that if the previous day has excess amount, the excess amount also incurs transportation costs because it will be transported again. For simplicity, we only consider the transportation cost from warehouse to district.
- Finally, there is an equity constraint. For each district, we can compute their weekly satisfaction level s_j,

$$s_j = \frac{\text{Weekly Satisfied Demand of District } j}{\text{Weekly Total Demand of District } j}. \tag{1}$$

  Regulation requires that, the highest level $\bar{s} = \max_{j \in \mathcal{J}} s_j$ and the lowest level $\underline{s} = \min_{j \in \mathcal{J}} s_j$, should satisfy:

$$\bar{s} - \underline{s} \leq 10\%. \tag{2}$$

To summarize, this problem has the following structure:

- For the objective:
  - ordering cost,
  - disposal cost,
  - transportation cost from warehouse i to district j each day,
  - holding cost of each warehouse each day,
- For the constraint:
  - the dynamics of holding amount of each warehouse each day,
  - the transportation amount from warehouse to district considering plan $y_{d,i,j}$ and excess amount,
  - the definition of disposal amount,
  - the equity constraint,
  - physical constraints (e.g., nonnegativity of decision variables).
- Note: y is not a recourse decision, it should be determined together with x.

2 Tasks

This practice consists of two tasks, one mandatory and one optional.

2.1 Task 1: Static Case (Mandatory 100%)

To make the decisions x_i and $y_{d,i,j}$, the food bank need extra information. You are provided with historical weekly demand data covering 5 years. Each year consists of 52 weeks, starting from Monday, January 2, 2012. The data files are named j-`training.npy`, where j corresponds to a district, and each file has the shape $[5, 52, 7]$ (number of years, number of weeks, number of days). Your task is to formulate a stochastic programming model using Sample Average Approximation (SAA) and to solve it. Specifically:

- Determine the inventory management plan, including x and y. These decisions should be unique and optimal for the SAA problem, not for each week.
- An advanced solution strategy (Bender's Decomposition, Progressive Hedging, etc.) may be considered instead of solving the full-scale model directly.
- Summarize your modeling approach, solution strategy, and results in the provided `.ipynb` file. Your work will be evaluated by
  - (70%) Correctness of the model formulation and decision quality (cost, whether regulation is satisfied, the food waste rate).
  - (30%) Solution strategy and computational efficiency.
- The related cost parameters are also given in the `Submission.ipynb` file. We recommend cvxpy as the modeling language.

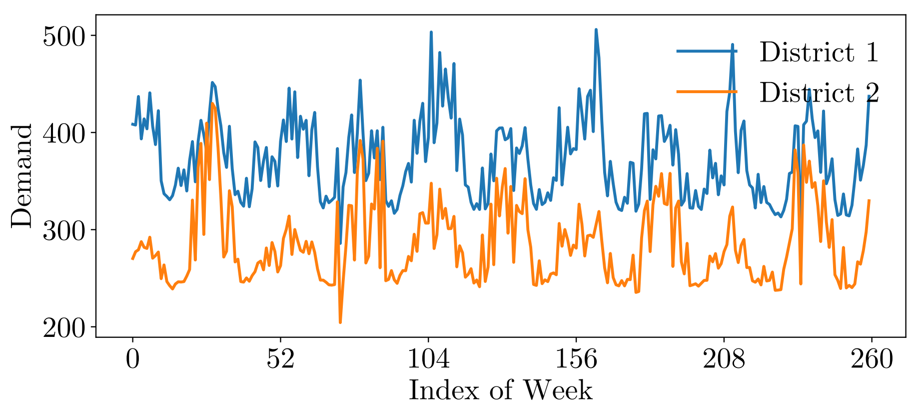

Figure 1: Monday demand for districts 1 and 2 over time.

2.2 Task 2: Dynamic Case (Optional Bonus)

Figure. 1 illustrates the Monday demand for districts 1 and 2 across all weeks in the dataset. The demand clearly exhibits temporal dependency and seasonality.

Consequently, the static SAA solution may be overly conservative during low-demand weeks, leading to excessive waste. To address this, a dynamic policy is required, where decisions are tailored to each week's characteristics.

Your task is to construct a policy that utilizes relevant temporal information to optimize decisions for specific weeks. Implement your policy to determine x, y for the 10th, 25th, and 43rd weeks of the 6th year (of which the demand data are hidden to you). Your decisions will be evaluated using the same metrics as in Task 1. Please document your approach to construct the policy in the same file.

To achieve this task, you may need extra reading materials regarding contextual decision-making. Please refer to [1] and [2] for a broader understanding. To guarantee the performance of your policy on unseen data, you may need to improve the robustness of your policy. For instance, if your policy contains some parameters, you may consider splitting your training data properly for cross-validation.

3 Submission

You will find a `submission.rar` file from the Edunao platform containing a directory of data, and a notebook used to formulate models and algorithms. Once you finish the task, keep the running log of the notebook and save required results.

To submit your group work, it is sufficient to upload the `.ipynb` file. Since this is a group work, we consider the following naming strategy for the submission.

- First, please find the `name_list.csv` file from the following link:

  `https://nextcloud.centralesupelec.fr/s/i6BgZ3mCJ8AyXtP`

- Second, check your index in this `name_list.csv`.
- Third, rename the `Submission.ipynb` by `index1-index2-...-indexN.ipynb`.
- For example, a group consists of Youssef Abouelaz, Elias Azar, and Xiaozhou Zhang will rename their file to `2-5-52.ipynb`.
- Lastly, upload the notebook using the previous link.

Please contact `xie.zhuojun@centralesupelec.fr` if you have any question regarding this group project.

Bibliography

[1] Sadana, U., Chenreddy, A., Delage, E., Forel, A., Frejinger, E., & Vidal, T. (2024). A survey of contextual optimization methods for decision-making under uncertainty. European Journal of Operational Research, 320(2), 271-289.

[2] Bertsimas, D., & Kallus, N. (2020). From predictive to prescriptive analytics. Management Science, 66(3), 1025-1044.